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App Launch Plan

For Project Three, I continued developing my Weight Tracker mobile application using Android Studio and Java. The purpose of this application is to help users monitor their weight, record their progress, and work toward personal weight goals. My goal was to create an application that is easy to navigate while providing useful features for managing weight information. I also wanted to maintain the visual design I created in Project Two, including the light pink background, colorful buttons, and simple screen layouts.

The application includes a login and account creation screen, a weight-tracking dashboard, and a notification settings screen. Users can create an account, log in, enter their weight and date, and save a goal weight. The dashboard displays recent weight entries and provides options to edit or delete saved information. I used SQLite to store user accounts, weight records, and goals locally on the device. This allows the application to retrieve saved information when users return.

One of the most important parts of developing this application was connecting the user interface to the Java code and database. In Project Two, I focused mainly on designing the screens and organizing the features. For Project Three, I worked on making those features functional. I created database methods to add, retrieve, update, and delete weight records. These operations allow users to manage their information rather than simply viewing a static design. I

also implemented account creation and login functionality. User passwords are stored using salted password hashes instead of plain text, which provides an additional layer of protection for account information. The application uses the logged-in user's account ID to retrieve the appropriate weight records and goal information. Another feature I added was goal tracking. Users can save a goal weight, and the application compares new weight entries with that goal. When an entered weight is equal to or below the saved goal, the application displays a congratulatory message. This feature helps users recognize their progress and gives them additional motivation to continue using the application. The notification settings screen allows users to change preferences for reminders, tips, progress updates, and SMS notifications. These settings are saved locally. However, actual SMS message delivery and automated reminders were not completed in this version. The application includes an in-app goal notification, but additional development would be required to send text messages.

Throughout development, I used Android Studio's build tools to check for coding errors and confirm that the project was compiled successfully. I also reviewed the connections between the XML layouts, Java classes, and database methods. When I encountered errors, I corrected the code and rebuilt the project to verify that the changes were accepted. Although the project was built successfully, I was unable to complete testing on an Android emulator or physical Android device. Because of this limitation, I cannot confirm that every feature works correctly during actual use. Before releasing the application to users, I would need to test account creation, login, weight entry, editing, deletion, goal tracking, and notification preferences. I would also test the layouts on different screen sizes to make sure the buttons and text remain accessible. Before launching the application, I would review its security, usability, and reliability. This would include confirming that account information is protected, checking how the application handles

incorrect input, and making sure users can navigate between screens without confusion. I would also complete SMS functionality, obtain the appropriate permissions, and test the notification process before advertising text-message alerts as an available feature.

If I continued developing this application, I would add a complete notification system, improve how weight history is displayed, and include charts to help users visualize their progress over time. I would also consider adding options to customize goals and reminders based on individual preferences.

This project helped me better understand the difference between designing a mobile application and developing a functional one. I gained more experience connecting Java code to XML layouts, working with SQLite databases, and implementing create, read, update, and delete operations. I also learned how important testing is before launching an application. Overall, Project Three gave me valuable experience with the mobile development process and showed me how individual features work together to create a complete application.